PSYCHOMETRIKA—VOL. 2, NO. 3
SEPTEMBER, 1937

MATHEMATICAL BIOPHYSICS OF CONDITIONING

N. RASHEVSKY
The University of Chicago

It is shown that from a somewhat more precise form of the fundamental postulates, used in a previous paper as a starting point for a system of mathematical biophysics of psychological phenomena, a mechanism can be derived, which represents the important features of conditioning.

In a previous paper, published in this journal and hereinafter referred to as I[1], we have outlined a general program for the development of mathematical biophysics in psychology. In a subsequent paper, referred to as II,[2] some particular points of the first, concerning conditioned reflexes were developed more in details. However in both those papers, while it was shown, that phenomena of conditioning are to be considered only as a particularly complex case of ordinary physical hysteresis,[3] properties of the mechanism, responsible for conditioning were not derived from the fundamental postulates, but had to be introduced as special assumptions. In the present paper we propose to show that no such additional assumptions of a physical nature are necessary and that provided we make our fundamental postulates somewhat more precise, the phenomena of conditioning can be derived from those, plus an assumption of a purely hystological nature. We now restate first our fundamental assumptions.

I.

While a continuous excitation of a nerve-fiber by means of a constant current usually results in a release of a simple excitation impulse, the qualitative and quantitative nature of which is entirely independent of the stimulus, provided the latter is strong enough to exceed the threshold, the situation for physiological stimuli, such as pressure, light, etc. is indifferent. Studies by Adrian and others show that a continuous stimulus sets off a volley of nerve-impulses following each other at approximately equal intervals. The stronger the

Edgar Adrian, Nobel prize 1932 "All or none law"

[1] N. Rashevsky: *Psychometrika*, **1**, 1, 1936.
[2] N. Rashevsky: *Psychometrika*: **1**, 265, 1936.
[3] N. Rashevsky: *ZS. f. Phys.* **53**, 102, 1929; *Jl. Gen. Psych*, **5**, 207, 1931; **5**, 368, 1931.

stimulus the shorter those intervals, or the higher the "frequency" of the sequence of impulses. While each individual impulse may be entirely independent of the intensity of the stimulus, preserving thus the "all or none law", yet the phenomenon as a whole gives a graded response to a graded stimulus. For not too strong intensities of the external stimulus the frequency of the volleys is approximately proportional to the intensity of the stimulus. We shall introduce the concept of the intensity of excitation E of a fiber, defining E as a quantity proportional to the frequency ν of impulses and to the intensity I of each individual impulse, Thus

$$E = I\,\nu. \tag{1}$$

If the "all or none law" holds for all fibers, then I is independent of the intensity S of the external stimulus and is a constant, characteristic of the fiber. ν however is proportional to S, but since S must in general exceed a threshold h in order that excitation would be released at all, ν is zero for $S = h$. Hence

$$\nu = a(S - h) \tag{2}$$

where however the factor of proportionality a may vary from fiber to fiber. (1) and (2) give

$$E = a\,I\,(S - h). \tag{3}$$

From physiological considerations it follows that (2) can at best be only an approximation. The interval between two successive impulses cannot be smaller than the refractory time θ of the fiber. Hence

$$\nu < \frac{1}{\theta} \tag{4}$$

no matter how strong S. As S increases indefinitely, ν must asymptotically tend to the value $\frac{1}{\theta}$, while for small values of S equation (2) holds approximately. There is of course an infinite number of functions which satisfy this requirement. Inasmuch as we have no empirical data yet to guide our choice, we shall choose the simplest possible one, namely:

$$\nu = \frac{1}{\theta}\,[1 - e^{-a\theta(S-h)}]. \tag{5}$$

For small values of $(S - h)$ we have

$$e^{-a\theta(S-h)} = 1 - a\theta\,(S - h)$$

and therefore (5) reduces to (2).

Using instead of (1) and (2) now (1) and (5) we obtain

$$E = \frac{I}{\theta}\left[1 - e^{-a\theta(S-h)}\right]. \tag{6}$$

With increasing S, E tends to I/θ.

We shall now consider the mechanism of transmission of excitation or the mechanism of inhibition. We assume that at the end of every excitatory fiber an *excitatory factor* ε is produced, according to the equation

$$\frac{d\varepsilon}{dt} = AE - a\varepsilon \tag{7}$$

where A and a are positive constants. We do not give any particular physical interpretation to this factor ε. It may be a special substance secreted in the immediate neighborhood of the end of the axon. Or it may be some other physico-chemical quantity, which follows *approximately* equ. (7). The development of the consequences of (7) is quite independent of such assumptions. In the future, of course, a physical interpretation of all equations, introduced here formally, such as (6) or (7) is to be attempted. For an inhibitory fiber we assume similarly, that its end produces an inhibitory factor j, according to

$$\frac{dj}{dt} = BE - bj \tag{8}$$

where B and b are positive constants.

For constant E, ε and j increase according to

$$\varepsilon = \frac{A}{a}\left(1 - e^{-at}\right); \quad j = \frac{B}{b}\left(1 - e^{-bt}\right)$$

(c.f. I) and tend asymptotically to $\frac{A}{a}$ and $\frac{B}{b}$ respectively. $\frac{A}{a}E$ represents the maximum value, that ε can reach for a continuous excitation of strength E.

From (7) and (8) it follows, that when for a long time the fibers are not excited at all, the values of ε and j are appreciably zero. For in case $E = 0$ we have

$$\frac{d\varepsilon}{dt} = -a\varepsilon \quad \text{and} \quad \frac{dj}{dt} = -bj$$

which gives upon integration

$$\varepsilon = \varepsilon_0 e^{-at} \quad ; \quad j = j_0 e^{-bt}$$

ε_0 and j_0 being some initial values. Both ε and j tend to zero in the absence of excitation of corresponding fibers. Since A, B and E are always positive, ε and j are also always positive.

We may consider a more general case, in which an excitatory as well as an inhibitory fiber both produce ε and j. Only an excitatory fiber always produces an excess of ε, an inhibitory fiber always an excess of j. For simplicity we shall consider here the above more restricted form of postulates.

Now we must specify the effect of ε and j on a neighboring, adjacent fiber, that lies in the vicinity of the end of the axon, which produces either ε or j. We shall postulate that whenever $\varepsilon > j$, $\varepsilon - j$ acts as an excitatory stimulus on any such adjacent fiber. That is if h_2 is the threshold of such an adjacent fiber, then if $\varepsilon - j > h_2$, this fiber becomes excited, with an intensity given by (3) or more generally by (6), in which $\varepsilon - j$ is substituted for S. For small $\varepsilon - j$ we thus have for the intensity E_2 of the adjacent fiber

$$E_2 = a_2 I_2 (\varepsilon - j - h_2). \tag{9}$$

When $\varepsilon - j < h_2$, and a fortiori when $\varepsilon < j$, no excitation occurs.

II.

Let us now consider the following structure, which is suggested by numerous neurological observations. An arrangement of neurones, forming a "closed circuit", as represented in Fig. 1, has frequently been observed and described. In this arrangement a neurone is stimulated by another and the latter in turn stimulates the first neurone. Various possible significances of such an arrangement have been discussed. Let us consider it from the point of view which interests us.

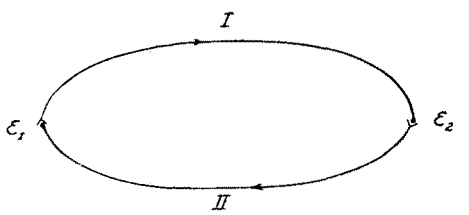

FIGURE 1

Let the threshold of I (Fig. 1) be h_1, that of II — h_2. In the absence of any external stimulation, none of the two neurones will be excited. Let however at the left end an amount of ε_1 be present, such

that $\varepsilon_1 > h_1$. This will produce a finite intensity of excitation E_1 in I, which in its turn will result in a production of ε_2 at the right end. If $\varepsilon_1 - h_1$ is sufficiently small, E_1 also will be sufficiently small and therefore ε_2 will be less than h_2. Hence II will not be excited. Under those conditions, when ε_1 again acquires the value zero, E_1 will become zero. Everything returns to its original state of non-excitation. Let however ε_1 exceed h_1 by such a large amount, that E_1 and therefore also ε_2 would become so large, that $\varepsilon_2 > h_2$. Then II will also become excited with an intensity E_2, which will result in the production of additional ε at the left end. This in its turn will result in an increase of E_1 and therefore in an increase of ε_2. The latter again increases E_2 and the process would thus tend automatically to infinity, if the linear relations between intensity of stimulation and intensity of excitation would hold exactly. As we have seen in section I, actually E_1 and E_2 tend to upper limits $\frac{I_1}{\theta_1}$ and $\frac{I_2}{\theta_2}$. Therefore the above described "self energizing" process will also actually stop, when E_1 and E_2 cannot increase any further, in spite of an increase of ε. But if now we bring the initial ε back to zero, it is possible that the additional ε produced by E_2 will be large enough to maintain E_1 excited and the system will remain in a continuous state of excitation even in the absence of an external stimulus.

These rather crude general considerations are borne out by mathematical analysis. To simplify the problem, we shall assume, that both I and II are very short and that the velocity of propagation of the excitation in both of them is very large, so that the time t_p which it takes for any variation of intensity of excitation at one end to reach the other is very small. In fact we consider it to be so small, that during this time neither ε_1 nor ε_2 can change appreciably.

We have

$$\left.\begin{aligned} E_1 &= \frac{I_1}{\theta_1}\left[1 - e^{-a_1\theta_1(\varepsilon_1 - h_1)}\right] \\ E_2 &= \frac{I_2}{\theta_2}\left[1 - e^{-a_2\theta_2(\varepsilon_2 - h_2)}\right] \end{aligned}\right\} \tag{10}$$

which gives

$$\left.\begin{aligned} \frac{d\varepsilon_1}{dt} &= \frac{AI_2}{\theta_2}\left[1 - e^{-a_2\theta_2(\varepsilon_2 - h_2)}\right] - a\varepsilon_1 \\ \frac{d\varepsilon_2}{dt} &= \frac{AI_1}{\theta_1}\left[1 - e^{-a_1\theta_1(\varepsilon_1 - h_1)}\right] - a\varepsilon_2. \end{aligned}\right\} \tag{11}$$

The analytic solution of the nonlinear system (11) is not known. We shall therefore investigate its properties by a graphical method.

$$\frac{d\varepsilon_1}{dt} \geqq 0 \text{ when}$$

$$\frac{AI_2}{\theta_2}[1 - e^{-a\theta_2(\varepsilon_2 - h_2)}] - a\varepsilon_1 \geqq 0$$

or

$$\varepsilon_1 \lesseqgtr \frac{AI_2}{a\theta_2}[1 - e^{-a_2\theta_2(\varepsilon_2 - h_2)}]. \tag{12}$$

The equality sign in (12) gives the equation of a line, which is zero for $\varepsilon_2 = h_2$ and tends asymptotically to $\frac{AI_2}{a\theta_2}$ with increasing ε_2. It is represented by the full line in Fig. 2. For all points below that line $\frac{d\varepsilon_1}{dt} > 0$, for all points above it $\frac{d\varepsilon_1}{dt} < 0$.

Similarly, $\frac{d\varepsilon_2}{dt} \geqq 0$, when

$$\frac{AI_2}{\theta_1}[1 - e^{-a_1\theta_1(\varepsilon_1 - h_1)}] - a\varepsilon_2 \lesseqgtr 0$$

which may be written thus:

$$1 - e^{-a_1\theta_1(\varepsilon_1 - h_1)} \geqq \frac{a\theta_1}{AI_1}\varepsilon_2$$

or

$$e^{-a_1\theta_1(\varepsilon_1 - h_1)} \lesseqgtr 1 - \frac{a\theta_1\varepsilon_2}{AI_1} = \frac{AI_1 - a\theta_1\varepsilon_2}{AI_1} < 1.$$

Taking logarithms:

$$-a_1\theta_1(\varepsilon_1 - h_1) \lesseqgtr \log\frac{AI_1 - a\theta_1\varepsilon_2}{AI_1} < 0.$$

Hence

$$a_1\theta_1(\varepsilon_1 - h_1) \geqq \log\frac{AI_1}{AI_1 - a\theta_1\varepsilon_2} > 0$$

or finally

$$\varepsilon_1 \geqq h_1 + \frac{1}{a_1\theta_1}\log\frac{AI_1}{AI_1 - a\theta_1\varepsilon_2}. \tag{13}$$

The sign of equality in (13) represents a line, shown by the broken line in Fig. 2. For $\varepsilon_2 = 0$, $\varepsilon_1 = h_1 > 0$. As ε_2 increases, ε_1 increases

also. When $\varepsilon_2 = \frac{AI_1}{a\theta_1}$, the denominator of the log becomes zero, and $\varepsilon_1 = \infty$. For still larger values of ε_2, ε_1 has no real values.

For all points below that curve $\frac{d\varepsilon_2}{dt} < 0$; for all points above $\frac{d\varepsilon_2}{dt} > 0$.

If, as represented on Fig. 2 the two curves intersect at all for $\varepsilon_1 > 0$ and $\varepsilon_2 > 0$, then they intersect at two points G_1 and G_2. For values of ε_1 and ε_2 corresponding to these two points $d\varepsilon_1/dt = 0$ and

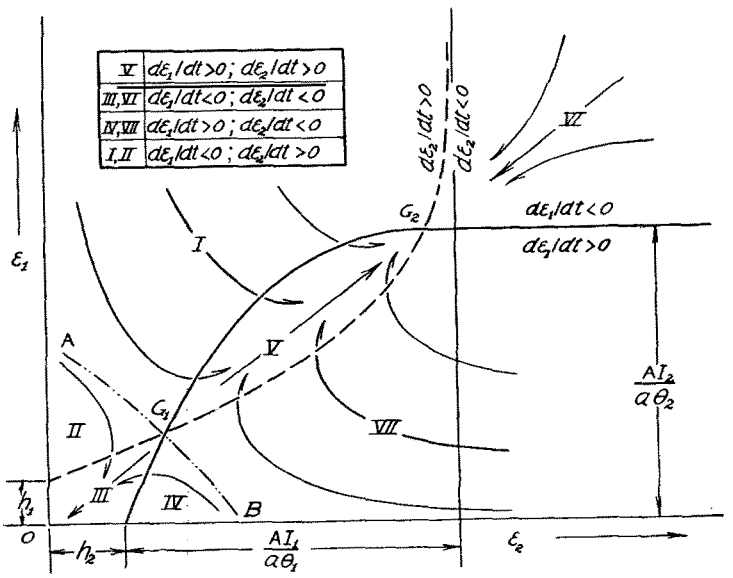

FIGURE 2

$d\varepsilon_2/dt = 0$. Hence the system does not change and is in equilibrium. It is however seen from inspection of Fig. 2 that while the configuration G_2 is stable, G_1 is unstable. Let the configurational point $(\varepsilon_1, \varepsilon_2)$ of the system be displaced from G_2 into region VI. Here as we see $d\varepsilon_1/dt < 0$ and $d\varepsilon_2/dt < 0$, therefore both ε_1 and ε_2 will decrease until they reach G_2 as shown by the arrows. If the point $(\varepsilon_1, \varepsilon_2)$ is in region V, then $d\varepsilon_1/dt > 0$ and $d\varepsilon_2/dt > 0$, the system moves again to G_2. In region I $d\varepsilon_1/dt < 0$, but $d\varepsilon_2/dt > 0$; the configurational point moves as indicated by the arrows until it comes into region V where it moves as we have seen to G_2. In region VII $d\varepsilon_1/dt > 0$, $d\varepsilon_2/dt < 0$ and we

have a similar situation. Fig. 2 indicates clearly, that while for any *small* displacement from G_2 the system returns to G_2 for any *small* displacement from G_1 it will move either to $\varepsilon_1 = \varepsilon_2 = 0$, or to G_2. The only exception is for displacements along the line AB, for which the system does return to G_1.

The above results can be demonstrated analytically by expanding the right-hand sides of (8) around G_1 and G_2, and keeping only the linear terms. We then obtain in the immediate vicinity of G_1 and G_2 for ε_1 and ε_2 a system of ordinary linear equations, the stability of whose solutions is determined and studied in the usual way. In this way it is also proven that the slope of the line AB at the point G_1 is equal to $-\sqrt{a_1 I_1 / a_2 I_2}$. This analytical method will be described elsewhere.

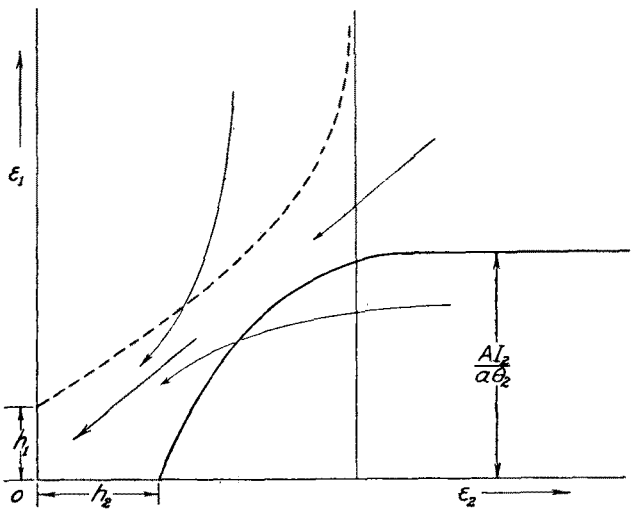

FIGURE 3

Thus the system considered has in this case in the absence of any external stimulation, two stable states of equilibrium. One corresponds to $\varepsilon_1 = \varepsilon_2 = 0$, the other to $\varepsilon_1 = \varepsilon_{0_1}$; $\varepsilon_2 = \varepsilon_{0_2}$. Then there is a line AB of unstable equilibrium, dividing the two states. As soon as by any external disturbance the system which was originally in a state $\varepsilon_1 = \varepsilon_2 = 0$ is brought into a state represented by a point to the right of AB, it "tips over" into G_2 $(\varepsilon_{0_1}, \varepsilon_{0_2})$ and remains there after the removal of the external disturbance.

If, as represented on Fig. (3) the full and broken lines do not

intersect at all, then in the absence of any external stimulation the only stable state is $\varepsilon_1 = \varepsilon_2 = 0$. Whether we shall have the case of Fig. (2) or that of Fig. (3) depends merely on the numerical values of the constants involved in our equations. The physical meaning of the case represented in Fig (3) is that the limiting value $\frac{I_1}{\theta_1}$ of E_1 is so small that even when it is reached, $\varepsilon_2 = \frac{A}{a}\frac{I_1}{\theta_1}$ is still less than h_2 and II therefore does not get excited, or that the limiting value $\frac{I_2}{\theta_2}$ of E_2 is too small to excite I.

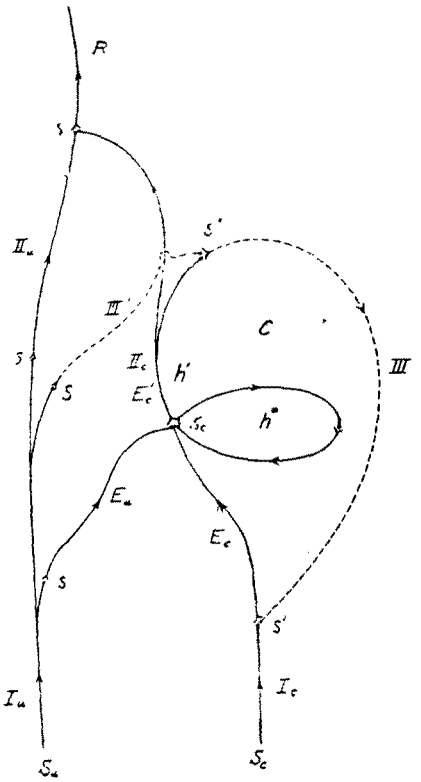

FIGURE 4

III.

Let us consider now a structure represented on Fig 4. Let the stimulus S_u produce an excitation in the nerve I_u with a threshold h_u, which carried over a number of synapses s finally results in some

reflex or reaction E. Let another series of nerves, I_c with threshold h_c, stimulated by a different stimulus S_c, lead to the synapse s_c at which it connects with a nerve II_c, which when excited also produces R through a final common path. Let this synapse s_c be excited also by a collateral of I_u and let it also connect to one end of a circuit, such as studied in the preceding section. Let that circuit be in an unexcited state and let the external excitation, which is necessary to bring it into an excited state, be $h^* \cdot h^*$ then represents the hysteresis threshold of this circuit. Let h' be the threshold of II_c and let ε_0 be the value of ε at the left end of the circuit, when it is in the stable excited state. Let furthermore the limiting vaues $\frac{I_u}{\theta_u}$ of E_u and $\frac{I_c}{\theta_c}$ of E_c be such that

$$P\frac{I_c}{\theta_c} < h' \quad ; \quad P = \frac{A}{a} \tag{14}$$

and

$$P\frac{I_c}{\theta_c} < h^* \quad ; \quad P\frac{I_u}{\theta_u} < h^* \tag{15}$$

but that

$$P\left(\frac{I_u}{\theta_u} + \frac{I_c}{\theta_c}\right) > h^* \tag{16}$$

and

$$P\frac{I_c}{\theta_c} + \varepsilon_0 > h'. \tag{17}$$

If a stimulus $S_u > h_u$ is applied, the reflex R is produced. However a stimulus S_c no matter how strong does not under those conditions produce R on account of (14), since PE_c is the maximum value that ε ever reaches for a continuous constant excitation E_c, and $\frac{I_c}{\theta_c}$ is the maximum possible value of E_c. If however both S_u and S_c are applied simultaneously and kept continuous for a sufficient time, then the amount ε and s_c will be

$$PE_u + PE_c = P(E_u + E_c).$$

On account of (16), this will become larger than h^*, if S_u and S_c are sufficiently strong, so that E_u and E_c are close enough to their limiting values I_u/θ_u and I_c/θ_c.

This will bring the circuit C into an excited state, in which in the absence of any external stimuli an amount ε_0 of ε will be produced

at the synapse s_c. When now again S_c is applied *alone* the total amount of ε at the synapse s_u will be for a continuous stimulation

$$P\frac{I_c}{\theta_c}+\varepsilon_0$$

and this is on account of (17) enough to excite II_c and therefore to produce a reflex R. Because of (15) application of S_u alone will not bring C into an excited state. The simultaneous application of S_u and S_c is essential.

We have here some of the principal features of Pavlov's conditioned reflex. In this simple scheme many fundamental details are however still missing.

We obtain some new features by complicating somewhat our scheme. If for instance on the efferent side of the circuit C the fiber II_c gives off a collateral, which through a synapse s'' excites an inhibitory fiber III, which ends at the synapse s', then after s_c has become conducting, a continuous stimulation of I_c produces a gradual increase of j at s' and thus inhibits the conditioned response. We have here the elements of internal inhibition of a conditioned reflex. If the fiber II_u sends off through a collateral an inhibitory fiber III' to the synapse s'', which inhibits the excitation of the inhibitory fiber III, then a simultaneous stimulation of I_u and I_c does not result in an inhibition of s'.

By integrating equ. (7) and (8), which govern the variation of ε and j at s_c and at other synapses for various types of stimulations a number of quantitative relations concerning the increase of the response R with the number of repetitions of simultaneous stimulations of I_u and I_c and other relations, may be derived along the lines indicated in I. The discussion of those relations must be reserved for a separate publication.